```
! pip install statsforecast
! pip install utilsforecast
```

```
Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from statsforecast) (3.1.2)
Collecting coreforecast>=0.0.12 (from statsforecast)
  Downloading coreforecast-0.0.17-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (3.6 kB)
Requirement already satisfied: numba>=0.55.0 in /usr/local/lib/python3.12/dist-packages (from statsforecast) (0.60.0)
Requirement already satisfied: numpy>=1.21.6 in /usr/local/lib/python3.12/dist-packages (from statsforecast) (2.0.2)
Requirement already satisfied: pandas>=1.3.5 in /usr/local/lib/python3.12/dist-packages (from statsforecast) (2.2.2)
Collecting scipy<1.16.0,>=1.7.3 (from statsforecast)
  Downloading scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 62.0/62.0 kB 2.9 MB/s eta 0:00:00
Requirement already satisfied: statsmodels>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from statsforecast) (0.14.6)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from statsforecast) (4.67.3)
Collecting fugue>=0.8.1 (from statsforecast)
  Downloading fugue-0.9.7-py3-none-any.whl.metadata (18 kB)
Collecting utilsforecast>=0.1.4 (from statsforecast)
  Downloading utilsforecast-0.2.15-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: threadpoolctl>=3 in /usr/local/lib/python3.12/dist-packages (from statsforecast) (3.6.0)
Collecting triad>=1.0.1 (from fugue>=0.8.1->statsforecast)
  Downloading triad-1.0.2-py3-none-any.whl.metadata (6.6 kB)
Collecting adagio>=0.2.6 (from fugue>=0.8.1->statsforecast)
  Downloading adagio-0.2.6-py3-none-any.whl.metadata (1.8 kB)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.55.0->stat
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.5->statsfore
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.5->statsforecast) (202
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.5->statsforecast) (2
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels>=0.13.2->statsforecast
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from statsmodels>=0.13.2->statsforec
Requirement already satisfied: narwhals>=2.0 in /usr/local/lib/python3.12/dist-packages (from utilsforecast>=0.1.4->statsforeca
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.3.5
Requirement already satisfied: pyarrow>=6.0.1 in /usr/local/lib/python3.12/dist-packages (from triad>=1.0.1->fugue>=0.8.1->stat
Requirement already satisfied: fsspec>=2022.5.0 in /usr/local/lib/python3.12/dist-packages (from triad>=1.0.1->fugue>=0.8.1->st
Downloading statsforecast-2.0.3-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (354 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 354.6/354.6 kB 7.3 MB/s eta 0:00:00
Downloading coreforecast-0.0.17-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (348 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 348.2/348.2 kB 20.6 MB/s eta 0:00:00
Downloading fugue-0.9.7-py3-none-any.whl (281 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 281.0/281.0 kB 20.6 MB/s eta 0:00:00
Downloading scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 37.3/37.3 MB 19.8 MB/s eta 0:00:00
Downloading utilsforecast-0.2.15-py3-none-any.whl (40 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 40.3/40.3 kB 3.0 MB/s eta 0:00:00
Downloading adagio-0.2.6-py3-none-any.whl (19 kB)
Downloading triad-1.0.2-py3-none-any.whl (59 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 59.9/59.9 kB 4.5 MB/s eta 0:00:00
Installing collected packages: scipy, coreforecast, utilsforecast, triad, adagio, fugue, statsforecast
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.3
    Uninstalling scipy-1.16.3:
      Successfully uninstalled scipy-1.16.3
Successfully installed adagio-0.2.6 coreforecast-0.0.17 fugue-0.9.7 scipy-1.15.3 statsforecast-2.0.3 triad-1.0.2 utilsforecast
Requirement already satisfied: utilsforecast in /usr/local/lib/python3.12/dist-packages (0.2.15)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from utilsforecast) (2.0.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from utilsforecast) (26.0)
Requirement already satisfied: pandas>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from utilsforecast) (2.2.2)
Requirement already satisfied: narwhals>=2.0 in /usr/local/lib/python3.12/dist-packages (from utilsforecast) (2.16.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.1->utilsfore
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.1->utilsforecast) (202
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.1->utilsforecast) (2
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.1.1
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from utilsforecast.plotting import plot_series
from utilsforecast.evaluation import evaluate
from utilsforecast.losses import *

import warnings
warnings.filterwarnings("ignore")
```

```
df = pd.read_csv('/content/daily_sales_french_bakery.csv', parse_dates=["ds"])
df.groupby('unique_id').filter(lambda x: len(x) > 28)
df = df.drop(['unit_price'], axis=1)
df.head()
```

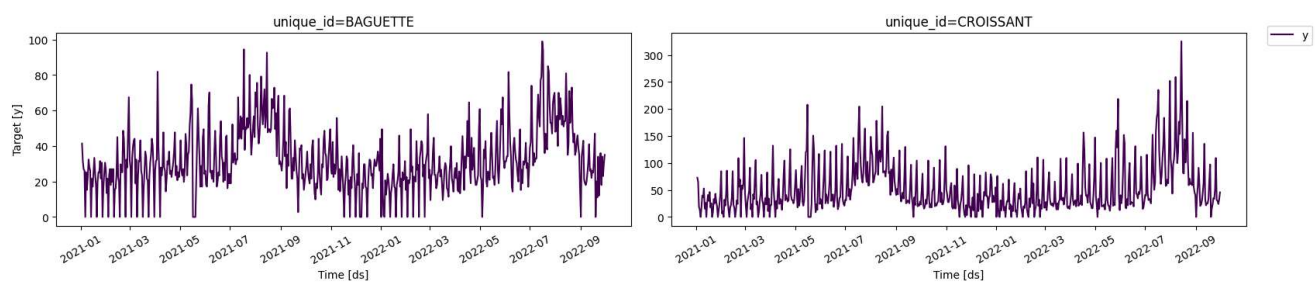
	unique_id	ds	y
0	12 MACARON	2022-07-13	10.0
1	12 MACARON	2022-07-14	0.0
2	12 MACARON	2022-07-15	0.0
3	12 MACARON	2022-07-16	10.0
4	12 MACARON	2022-07-17	30.0

Next steps: Generate code with df New interactive sheet

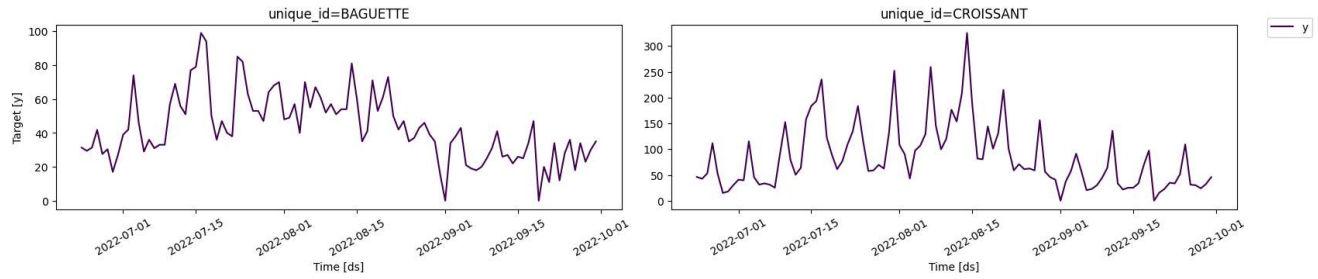
```
df['unique_id'].unique()
```

```
array(['12 MACARON', 'ARMORICAIN', 'ARTICLE 295', 'BAGUETTE',
       'BAGUETTE APERO', 'BAGUETTE GRAINE', 'BANETTE', 'BANETTINE',
       'BOISSON 33CL', 'BOTTEREAU', 'BOULE 200G', 'BOULE 400G',
       'BOULE POLKA', 'BRIOCHE', 'BRIOCHE DE NOEL', 'BRIOCHETTE',
       'BROWNIES', 'BUCHE 4PERS', 'BUCHE 6PERS', 'BUCHE 8PERS',
       'CAFE OU EAU', 'CAKE', 'CAMPAGNE', 'CARAMEL NOIX',
       'CEREAL BAGUETTE', 'CHAUSSON AUX POMMES', 'CHOCOLAT',
       'CHOU CHANTILLY', 'COMPLET', 'COOKIE', 'COUPE', 'CROISSANT',
       'CROISSANT AMANDES', 'CRUMBLE', 'CRUMBLECARAMEL OU PISTAE',
       'DELICETROPICAL', 'DEMI BAGUETTE', 'DEMI PAIN', 'DIVERS BOISSONS',
       'DIVERS BOULANGERIE', 'DIVERS CONFISERIE', 'DIVERS PATISSERIE',
       'DIVERS SANDWICHS', 'DIVERS VIENNOISERIE', 'DOUCEUR D HIVER',
       'ECLAIR', 'ECLAIR FRAISE PISTACHE', 'ENTREMETS', 'FICELLE',
       'FINANCIER', 'FINANCIER X5', 'FLAN', 'FLAN ABRICOT',
       'FONDANT CHOCOLAT', 'FORMULE PATE', 'FORMULE PLAT PREPARE',
       'FORMULE SANDWICH', 'FRAISIER', 'FRAMBOISIER', 'GACHE',
       'GAL FRANGIPANE 4P', 'GAL FRANGIPANE 6P', 'GAL POIRE CHOCO 4P',
       'GAL POIRE CHOCO 6P', 'GAL POMME 4P', 'GAL POMME 6P',
       'GALETTE 8 PERS', 'GD FAR BRETON', 'GD KOUIGN AMANN', 'GD NANTAIS',
       'GD PLATEAU SALE', 'GRAND FAR BRETON', 'GRANDE SUCETTE',
       'GUERANDAIS', 'KOUIGN AMANN', 'MACARON', 'MERINGUE',
       'MILLES FEUILLES', 'MOISSON', 'NANTAIS', 'NID DE POULE',
       'NOIX JAPONAISE', 'PAILLE', 'PAIN', 'PAIN AU CHOCOLAT',
       'PAIN AUX RAISINS', 'PAIN BANETTE', 'PAIN CHOCO AMANDES',
       'PAIN DE MIE', 'PAIN GRAINES', 'PAIN NOIR', 'PAIN S/SEL',
       'PAIN SUISSE PEPITO', 'PALET BRETON', 'PALMIER', 'PARIS BREST',
       'PATES', 'PLAQUE TARTE 25P', 'PLAT', 'PLAT 6.50E', 'PLAT 7.00',
       'PLAT 7.60E', 'PLAT 8.30E', 'PLATPREPARE5.50', 'PLATPREPARE6.00',
       'PLATPREPARE6.50', 'PLATPREPARE7.00', 'PT NANTAIS',
       'PT PLATEAU SALE', 'QUIM BREAD', 'REDUCTION SUCREES 12',
       'REDUCTION SUCREES 24', 'RELIGIEUSE', 'ROYAL', 'ROYAL 4P',
       'ROYAL 6P', 'SABLE F  P', 'SACHET DE CROUTON',
       'SACHET DE VIENNOISERIE', 'SACHET VIENNOISERIE', 'SAND JB',
       'SAND JB EMMENTAL', 'SANDWICH COMPLET', 'SAVARIN', 'SEIGLE',
       'SPECIAL BREAD', 'SPECIAL BREAD KG', 'ST HONORE', 'SUCETTE',
       'TARTE FINE', 'TARTE FRAISE 4PER', 'TARTE FRAISE 6P',
       'TARTE FRUITS 4P', 'TARTE FRUITS 6P', 'TARTELETTE',
       'TARTELETTE CHOC', 'TARTELETTE COCKTAIL', 'TARTELETTE FRAISE',
       'THE', 'TRADITIONAL BAGUETTE', 'TRAITEUR', 'TRIANGLES',
       'TROIS CHOCOLAT', 'TROPEZIENNE', 'TROPEZIENNE FRAMBOISE', 'TULIPE',
       'VIENNOISE', 'VIK BREAD'], dtype=object)
```

```
plot_series(df=df, ids=['BAGUETTE', 'CROISSANT'], palette="viridis")
```




```
plot_series(df=df, ids=['BAGUETTE', 'CROISSANT'], max_insample_length=100 ,palette="viridis")
```



Baseline Models

```
from statsforecast import StatsForecast
from statsforecast.models import Naive, HistoricAverage, WindowAverage, SeasonalNaive
```

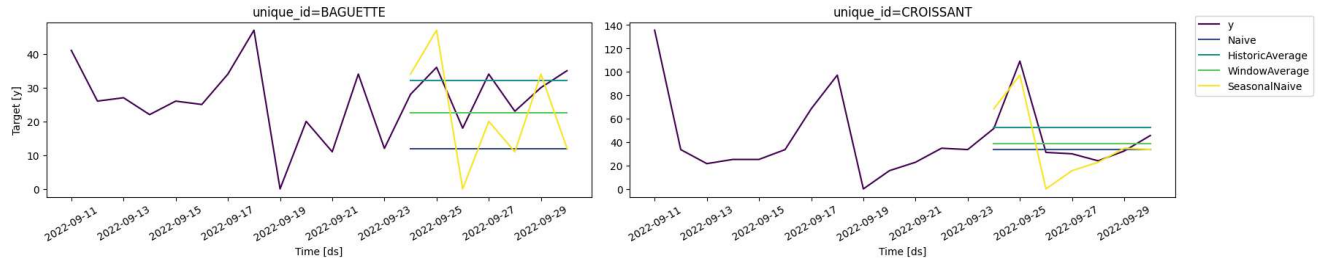
```
horizon = 7

models = [
    Naive(),
    HistoricAverage(),
    WindowAverage(window_size=7),
    SeasonalNaive(season_length=7)
]
```

```
test = df.groupby('unique_id').tail(7)
train = df.drop(test.index).reset_index(drop=True)
```

```
obj = StatsForecast(models=models, freq='D')
obj.fit(df=train)
preds = obj.predict(h=horizon)
eval_df = pd.merge(test, preds, 'left', ['ds', 'unique_id'])
```

```
plot_series(df=df,
            forecasts_df=preds,
            ids=['BAGUETTE', 'CROISSANT'],
            max_insample_length=20,
            palette='viridis')
```

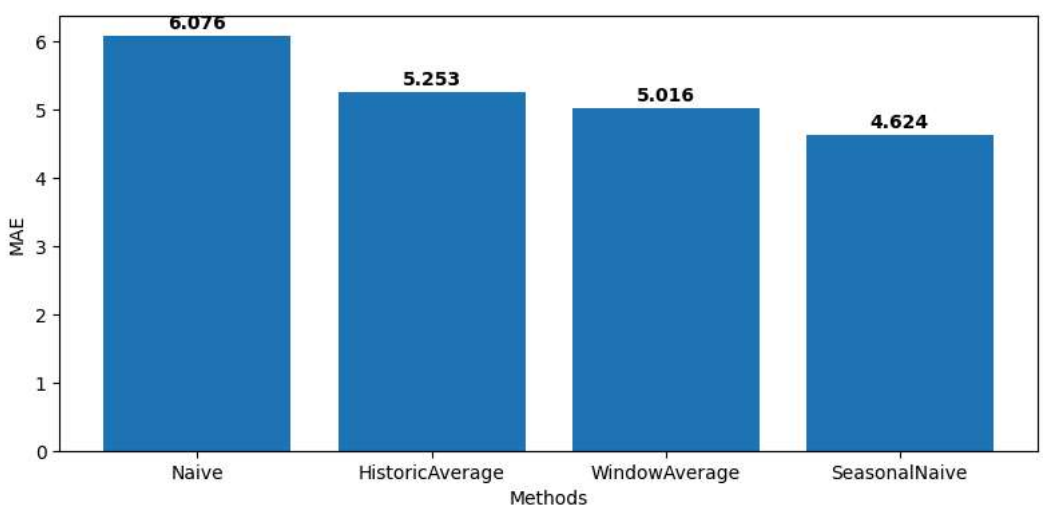



```
evaluation = evaluate(
    df = eval_df,
    metrics=[mae],
)
evaluation.head()
```

	unique_id	metric	Naive	HistoricAverage	WindowAverage	SeasonalNaive
0	12 MACARON	mae	2.857143	6.961771	3.469388	4.285714
1	ARTICLE 244	mae	NaN	NaN	NaN	NaN
2	ARTICLE 295	mae	NaN	NaN	NaN	NaN
4	BAGUETTE APERO	mae	0.000000	0.537572	0.642857	0.642857

Next steps: Generate code with evaluation New interactive sheet

```
evaluation = evaluation.drop(['unique_id'], axis=1).groupby('metric').mean().reset_index()
evaluation
```

	metric	Naive	HistoricAverage	WindowAverage	SeasonalNaive
0	mae	6.07604	5.253154	5.016269	4.624203

```
methods = evaluation.columns[1:].tolist()
values = evaluation.iloc[0, 1:].tolist()

plt.figure(figsize=(8, 4))
bars = plt.bar(methods, values)

for bar, value in zip(bars, values):
    plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.05,
             f'{value:.3f}', ha='center', va='bottom', fontweight='bold')

plt.xlabel('Methods')
plt.ylabel('MAE')
plt.tight_layout()
plt.show()
```